

Exercise Sheet 2

No Fear of Numbers: Introduction to Quantitative Data Analysis in R

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Univariate Analysis

In the previous exercise, we got to know our dataset and some of the main variables. In this sheet, we focus on univariate analysis looking at **one variable at a time**. Univariate analysis is the starting point of any data analysis. It helps us **understand distributions**, typical values, and strange values and is the basis for all further steps: bivariate relationships and regression models.

In the context of our research story, we are interested in **overall life satisfaction** (blcd01) and **having children (yes/no)** (from blcd06). Before we relate these outcomes to other factors, we first want to understand

- How are these variables distributed in our sample?
- How are other key variables (e.g. gender, age, burden, work–life compatibility) distributed?

In this exercise, you will:

- transform categorcial variables to factors
- describe nominal variables using absolute and relative frequencies, mode, and bar charts,
- describe ordinal variables using cumulative frequencies, median, percentiles,
- describe metric variables using mean, standard deviation, percentiles, boxplots, histograms, skewness and kurtosis, standard error, confidence intervals

0 Set-up: Load packages and dataset, load data frame

Install `tidyverse`, `haven`, `janitor`, `sjmisc`, `scales`, `skimr` and `moments` if not already installed and load these packages. Set the “Exercise” folder as your working directory and load dataset “02_qa_uni_data.sav” as a data frame called `mydata2`.

Code it (or faster: copy and paste the code) like this:

```

# Install tidyverse
if (!requireNamespace("tidyverse", quietly = TRUE))
  install.packages("tidyverse")

# Install haven
if (!requireNamespace("haven", quietly = TRUE))
  install.packages("haven")

# Install readxl
if (!requireNamespace("janitor", quietly = TRUE))
  install.packages("janitor")

# Install skimr
if (!requireNamespace("skimr", quietly = TRUE))
  install.packages("skimr")

# Install scales
if (!requireNamespace("scales", quietly = TRUE))
  install.packages("scales")

# Install moments
if (!requireNamespace("moments", quietly = TRUE))
  install.packages("moments")

# Install sjmisc
if (!requireNamespace("sjmisc", quietly = TRUE))
  install.packages("sjmisc")

## load packages
library(tidyverse)
library(haven)
library(sjmisc)
library(skimr)
library(janitor)
library(scales)
library(moments)

# set working directory
setwd(
  ↪ "C:/Users/Susanne/Nextcloud/share_DSC/003_Trainings/001_Workshops/2026/2026-03-11_SdV_MM_Quantitative_Data_Anal
  ↪ )

# Load the Nacaps data set and name this data frame "mydata2"
mydata2 <- read_sav("02_qa_uni_data.sav")

```

On the right side, in the Environment pane, the data frame `mydata2` should now appear under **Data**.

Exercise 1: Categorical variables: Children and gender

1.1. Calculate absolute and relative frequencies for `blcd06` children (1 = yes/ 2 = no).

Use the function `tabyl()` from the *janitor* package to create a frequency table for variable `blcd06`. Look at the counts and percentages in the output, what does it tell you?

Solution:

```
# calculate absolute and relative frequencies with tabyl function
tabyl(mydata2$blcd06)
```

```
## mydata2$blcd06      n      percent valid_percent
##              1  423 0.1796941      0.192623
##              2 1773 0.7531861      0.807377
##              NA  158 0.0671198           NA
```

tabyl reports the absolute frequencies (n), relative frequencies of all categories incl missings (percent, e.g. 0.75 = 75%) and relative frequencies excl. missings (valid_percent).

In our sample, 80% of doctoral researchers have no children (code 2), and 20% have at least one child.

1.2. Alternative: Calculate absolute and relative frequencies for adem01 gender *with labels*

tabyl() will show the **numeric codes**, which is not very intuitive. To make the table easier to read, we want to display the **variable and value labels** instead.

a. To do so, first convert adem01 into a *factor* with labels.

Use mutate() and adem01 = as_factor(adem01, levels="labels") to transform adem01 into a factor variable. Save the result again as mydata2. The code should look like this:

```
# convert the labelled variable into a factor whose levels are the value labels (e.g. "female",
  ↪ "male", ...)
mydata2 <- mydata2 %>%
  mutate(
    adem01 = as_factor(adem01, levels="labels")
  )
```

b. Create a frequency table for adem01 using the tabyl() function.

Solution:

```
# calculate absolute and relative frequencies with tabyl function
tabyl(mydata2$adem01)
```

```
## mydata2$adem01      n      percent valid_percent
##          female 1153 0.4898045879      0.4900127
##          male  1068 0.4536958369      0.4538887
##          other   132 0.0560747664      0.0560986
##          <NA>     1 0.0004248088           NA
```

1.3. Identify the *mode* of blcd06 and adem01.

The mode is the category that appears most often. As there are variables with little categories, we can easily identify it by looking at the frequency tables we created with tabyl().

Solution:

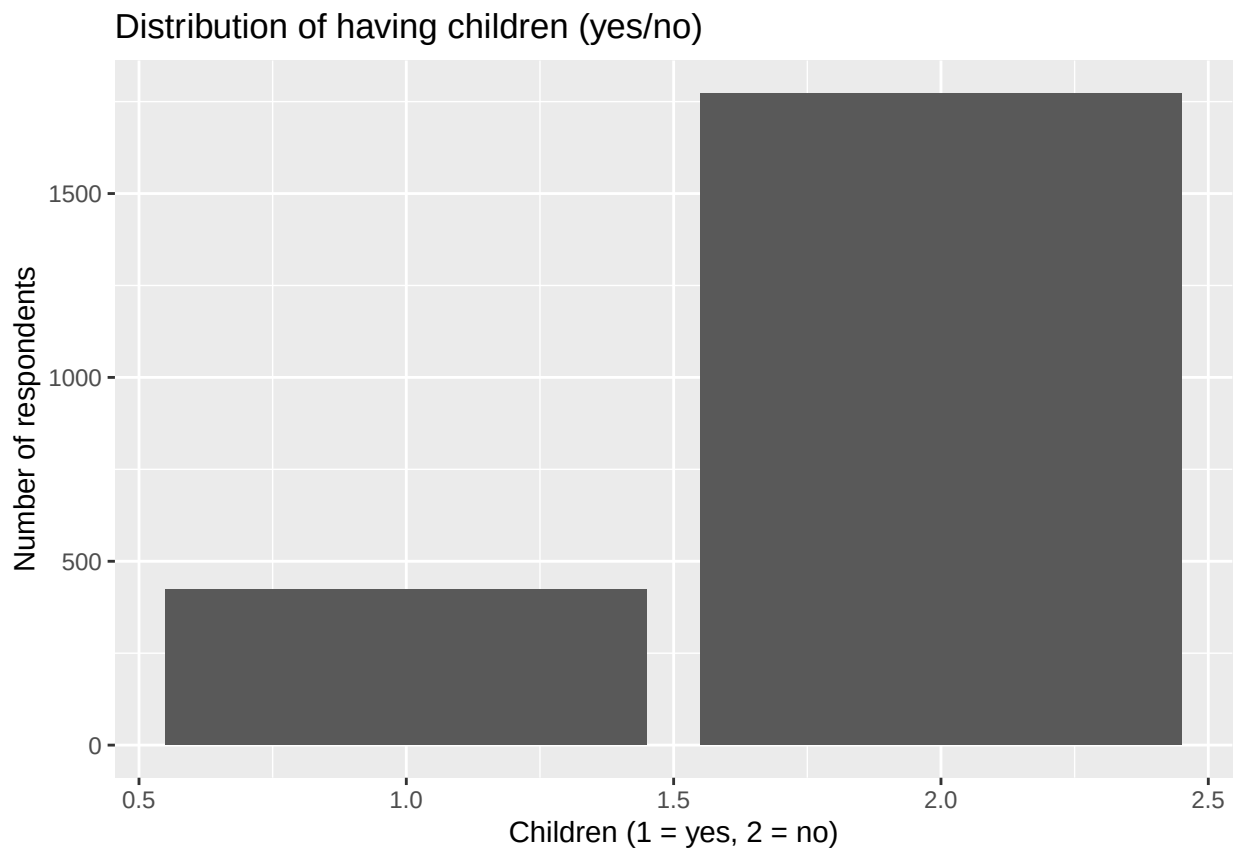
For gender adem01, the most frequent category is female. For blcd06, most people in the sample do not have children.

1.4. Visualize the distribution of blcd09 children (yes/no) in a bar chart.

Draw a bar chart with blcd06 on the x-axis using ggplot() function. The height of the bars should show how many people are in each category (1/2). The code should look like this:

```
mydata2 %>% # take mydata2 as input
  ggplot(aes(x = blcd06)) + # put blcd06 on x-axis
  geom_bar() + # draw bars, counting rows per category
  labs( # add labels and title
    x = "Children (1 = yes, 2 = no)", # label for x-axis
    y = "Number of respondents", # label for y-axis
    title = "Distribution of having children (yes/no)" # main title of the plot
  )
```

```
## Warning: Removed 158 rows containing non-finite outside the scale range
## (`stat_count()`).
```



What do you notice?

Solution:

The categories on the x-axis don't make any sense because R does not know that it should treat the variable like a categorical variable.

1.5. Create a similar bar chart for the distribution of gender adem01. What do you notice?

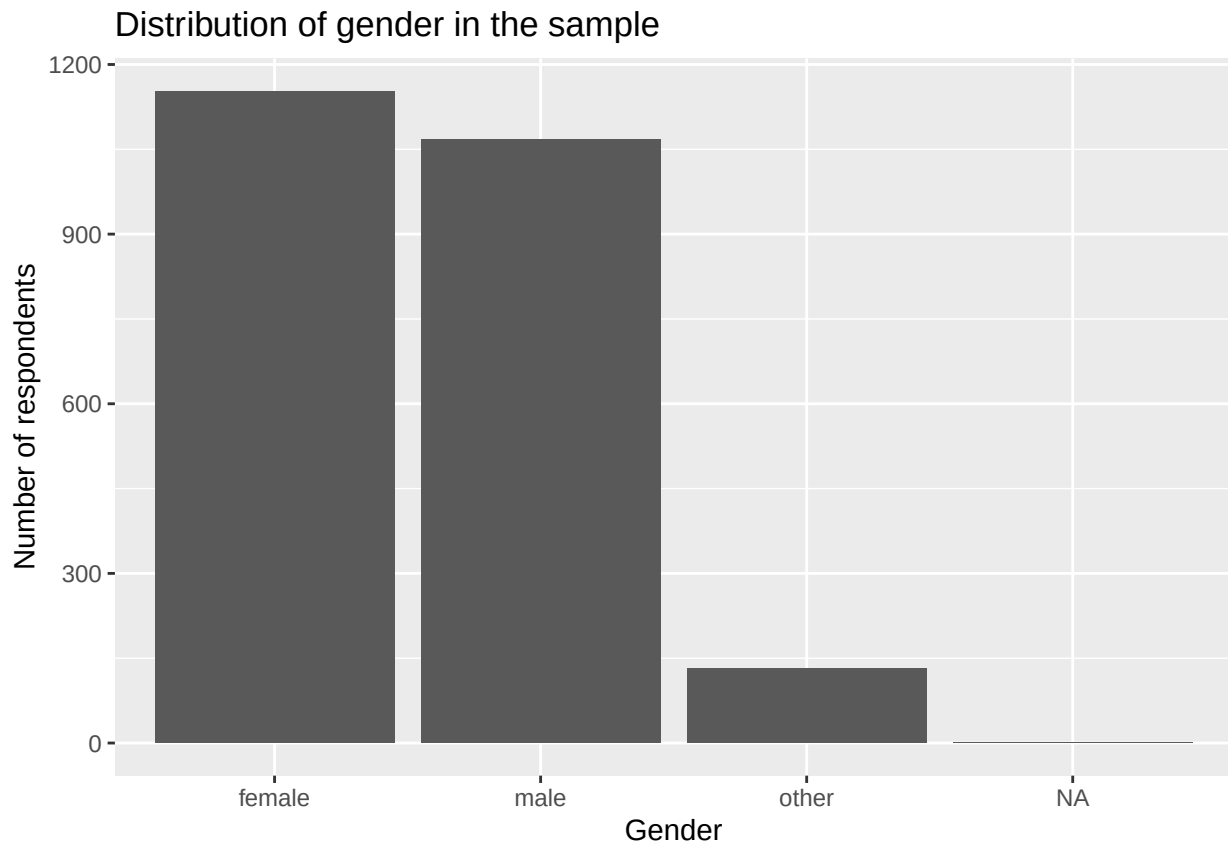
Solution:

```
mydata2 %>%
  ggplot(aes(x = adem01)) +
  geom_bar() +
  labs(
    x = "Gender",
```

```

y = "Number of respondents",
title = "Distribution of gender in the sample"
)

```



Because the variable is transformed into a factor, the categories on the axes make much more sense now.

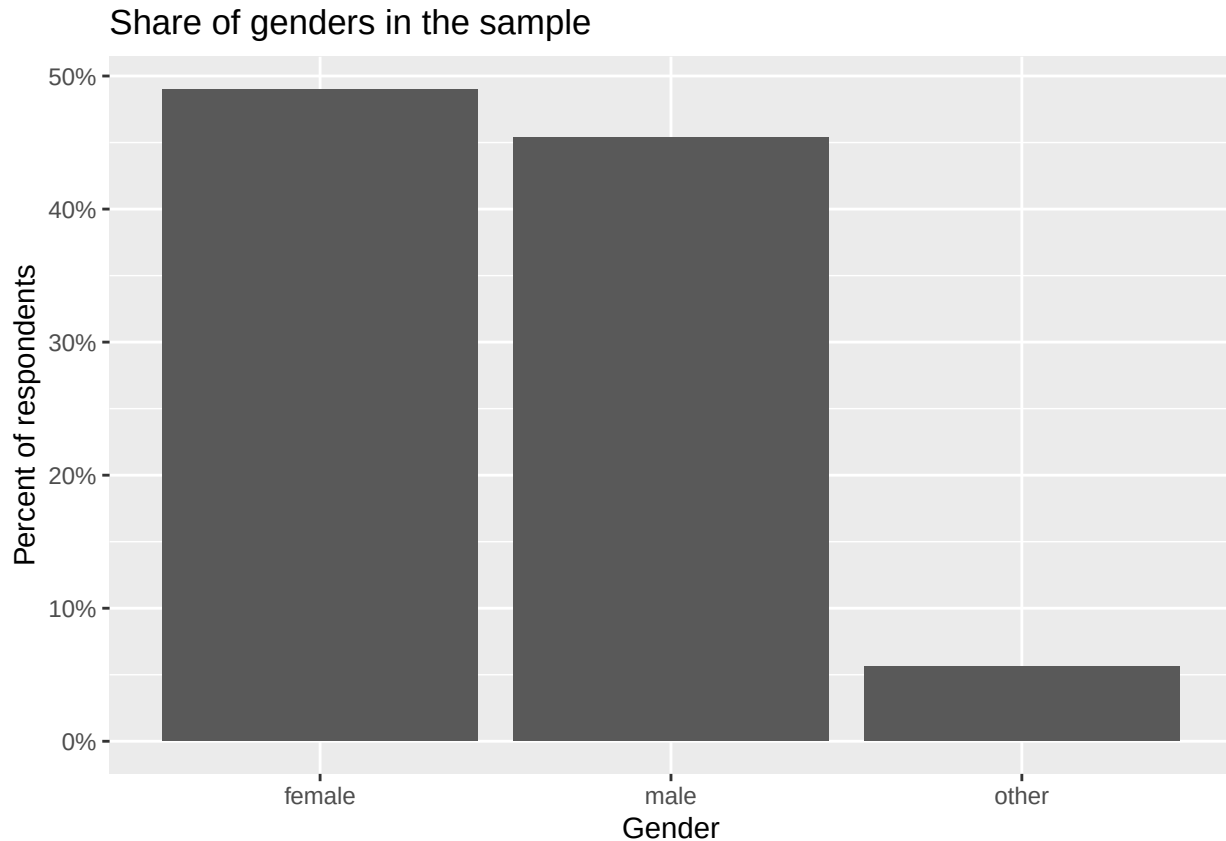
1.6. Modify the graph so 1.) missings do not appear and 2.) percentages are plotted.

To do so, you need to remove missing values out of `adem01`, count the cases in the category and use these information for plotting the bar chart. Modify the `ggplot()` function like this:

```

mydata2 %>%
  filter(!is.na(adem01)) %>%           # remove missing values of adem01
  count(adem01) %>%                   # count how many cases in each category
  mutate(prop = n / sum(n)) %>%      # compute proportion (relative frequency)
  ggplot(aes(x = adem01, y = prop)) + # map category to x-axis, proportion to y-axis
  geom_col() +                        # draw bars with given heights (prop)
  scale_y_continuous(labels = percent) + # scales package: show y-axis as percentages
  labs(
    x = "Gender",                     # x-axis label
    y = "Percent of respondents",     # y-axis label
    title = "Share of genders in the sample"
  )

```



Exercise 2: Metric variables: Overall life-satisfaction

Our outcome variable `bpsy01` measures the **overall life satisfaction** rated on a scale from 0 to 10. Technically, it is **an ordinal variable**, but in research practices, scales like this are often **treated as metric**. And this is what we are going to do.

2.1. Look at the distribution of `bpsy01`. Calculate the *absolute, relative and cumulative frequencies*, read the *percentiles* and look at the *mean and standard deviation* using the `frq()` function.

Solution:

```
# frequency table for bpsy01
frq(mydata2$bpsy01)
```

```
## Overall life satisfaction (x) <numeric>
## # total N=2354 valid N=2197 mean=7.04 sd=1.90
##
## Value | Label | N | Raw % | Valid % | Cum. %
## -----
## 0 | 0 not at all satisfied | 10 | 0.42 | 0.46 | 0.46
## 1 | 1 | 13 | 0.55 | 0.59 | 1.05
## 2 | 2 | 43 | 1.83 | 1.96 | 3.00
## 3 | 3 | 92 | 3.91 | 4.19 | 7.19
## 4 | 4 | 87 | 3.70 | 3.96 | 11.15
## 5 | 5 | 137 | 5.82 | 6.24 | 17.39
```

```
##      6 |                6 | 243 | 10.32 |    11.06 |    28.45
##      7 |                7 | 531 | 22.56 |    24.17 |    52.62
##      8 |                8 | 596 | 25.32 |    27.13 |    79.75
##      9 |                9 | 342 | 14.53 |    15.57 |    95.31
##     10 |    10 fully satisfied | 103 |  4.38 |     4.69 |   100.00
##    <NA> |                <NA> | 157 |  6.67 |     <NA> |     <NA>
```

You can use these **cumulative percentages** to read off percentiles for an metric variable.

In the `frq()` output, look for the column with cumulative (valid) percent. Then:

- The **25th percentile (P25)** is the first value where the cumulative percent is greater than or equal to 25%. In this case, it's value 6 on the scale.
- The **50th percentile (P50, median)** is the first value where the cumulative percent is 50%.
- The **75th percentile (P75)** is the first category where the cumulative percent is 75%.

In other words: move down the cumulative percent column until you reach or pass the percentile you are interested in. The value at that point is the percentile value.

2.2. Examine bpsy01 using the `summary` and the `skim()` function from the *skimr* package. What can you see here? How do these functions differ?

Solution:

```
# distribution using the summary function
summary(mydata2$bpsy01)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##    0.000   6.000   7.000   7.036   8.000   10.000    157
```

```
# distribution using the skim function
skim(mydata2$bpsy01)
```

Table 1: Data summary

Name	mydata2\$bpsy01
Number of rows	2354
Number of columns	1
Column type frequency:	
numeric	1
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
data	157	0.93	7.04	1.9	0	6	7	8	10	

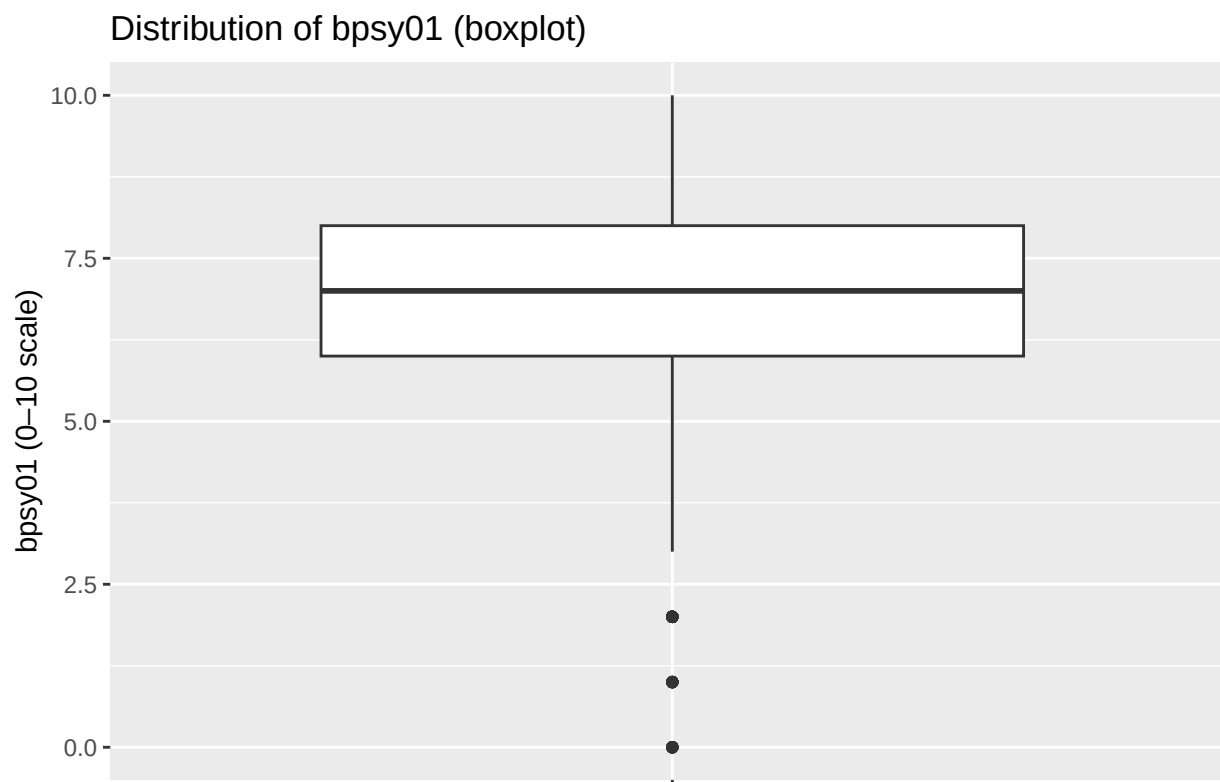
`summary()` is a base R function that gives a quick overview of each variable in a data frame (min, quartiles, median, mean, max for numeric variables; most frequent categories for factors; plus NAs).

skim() provides a more detailed, tidy overview of the data. It **detects variable types** and reports **type-specific statistics** (e.g. mean, SD, percentiles and a mini histogram for numeric variables; number of levels and top categories for factors).

2.3. To visualize the distribution of bpsy01, draw a boxplot with ggplot.

The code goes like this:

```
mydata2 %>%  
  filter(!is.na(bpsy01)) %>%           # remove missings  
  ggplot(aes(x = "", y = bpsy01)) +     # bpsy01 on y-axis  
  geom_boxplot() +                     # draw boxplot  
  labs(  
    x = "",  
    y = "bpsy01 (0-10 scale)",  
    title = "Distribution of bpsy01 (boxplot)"  
  )
```



What does the boxplot show you?

Solution:

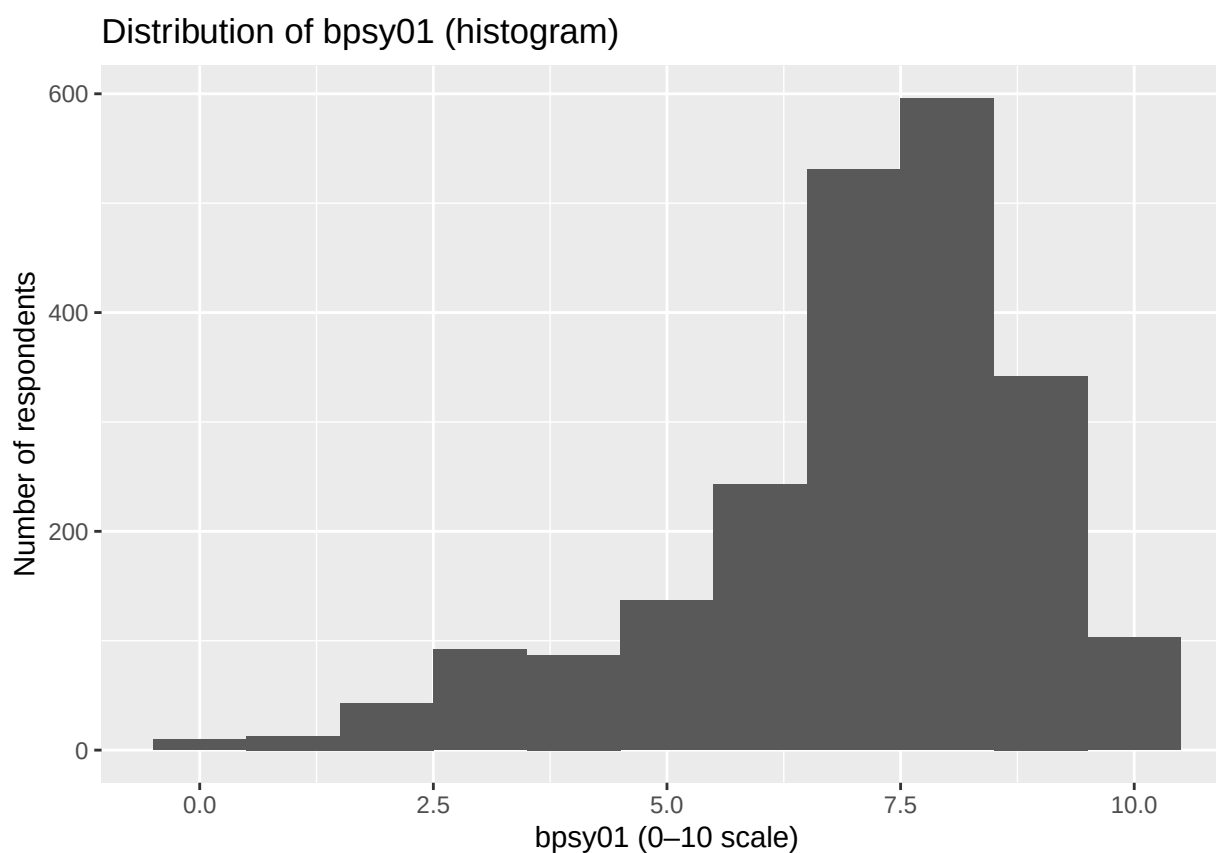
In a boxplot, the box shows the middle 50% of the values (between the 25th and 75th percentile), the horizontal line inside the box is the median, the whiskers show the typical range of the data, and the dots are outliers.

Here we see that most respondents have high values on bpsy01: the middle 50% lie roughly between 6 and 8, with a median around 7. A few respondents report much lower values (below 3), which appear as outliers at the bottom of the plot.

2.4. To see the overall shape of the distribution of `bpsy01`, create a *histogram* using `ggplot()`. Use `bpsy01` on the x-axis and choose a bin width of 1 to show the 0–10 scale clearly.

The code goes like this:

```
# histogram for bpsy01
mydata2 %>%
  filter(!is.na(bpsy01)) %>%           # remove missings
  ggplot(aes(x = bpsy01)) +            # bpsy01 on x-axis
  geom_histogram(binwidth = 1) +       # bins of width 1 (0,1,2,...,10)
  labs(
    x = "bpsy01 (0-10 scale)",
    y = "Number of respondents",
    title = "Distribution of bpsy01 (histogram)"
  )
```



What do you see regarding the skewness of `bpsy01`?

Solution:

The histogram shows that `bpsy01` is clearly concentrated at the higher end of the 0–10 scale: most respondents have values between about 6 and 9, with a peak around 7–8. Very low values (below 3) are rare, so the distribution is slightly **left-skewed (longer tail towards the lower scores)**.

Note: If you want to compute skewness and kurtosis, you can do it using the following functions from the *moments* package:

```
skewness(mydata2$bpsy01, na.rm = TRUE)
```

```
## [1] -1.051064
```

```
kurtosis(mydata2$bpsy01, na.rm = TRUE)
```

```
## [1] 4.064778
```

2.5. Let's look at inference statistics for bpsy01. Calculate *standard errors* and both *95 % confidence intervals* using the `summarise()` function.

Code it like this:

```
mydata2 %>%
  summarise(
    n      = sum(!is.na(bpsy01)),      # number of valid cases
    mean_val = mean(bpsy01, na.rm = TRUE), # mean
    sd_val  = sd(bpsy01, na.rm = TRUE),  # standard deviation
    se_val  = sd_val / sqrt(n),          # standard error of the mean
    ci_lower = mean_val - 1.96 * se_val,  # 95% CI lower bound
    ci_upper = mean_val + 1.96 * se_val  # 95% CI upper bound
  )
```

```
## # A tibble: 1 x 6
##       n mean_val sd_val se_val ci_lower ci_upper
##   <int>   <dbl> <dbl> <dbl>   <dbl>   <dbl>
## 1  2197     7.04  1.90  0.0406     6.96     7.12
```

Take Home checklist: Univariate analysis in R

Step	Question / task	Useful functions / tools
1	Do your categorical survey variables have readable categories (factors)?	Convert labelled variables with <code>as_factor()</code> , then use <code>tabyl()</code> or <code>ggplot()</code> on the factors.
2	Are missing values coded correctly?	Recode special codes to NA (e.g. <code>mutate(across(..., ~ ifelse(.x < 0, NA, .x)))</code>)
3	For categorical variables: how often does each category occur?	<code>tabyl()</code> , <code>count()</code> , bar charts with <code>ggplot(aes(x = var)) + geom_bar() / geom_col()</code>
5	For metric variables: how are responses distributed along the scale?	<code>frq()</code> , cumulative % and percentiles from the table
6	For metric variables: what are central tendency and spread and shape?	<code>mean()</code> , <code>sd()</code> , <code>var()</code> , <code>quantile()</code> <code>skmr()</code> , SE and CI via <code>summarise()</code> , box plot with <code>geom_boxplot()</code> , histogram with <code>geom_histogram()</code>
7	Are there obvious problems in the distributions?	Check tables, boxplots and histograms for impossible values, extreme outliers, very strong skewness etc.